Full Stack Development
B.Tech-VI Semester (20ITL601/SOC4)

Lectures	:	2 Hours / Week	Tutorial	:	0	Practical	:	3
CIE Marks	:	30	SEE Marks	:	70	Credits	:	3.5

Prerequisites:

Course Objectives:

The course aims to enable the students

- to learn React.js front end framework basics
- to learn React.js Application Testing, Deployment and Node.js basics
- to create web server using Express
- to work with Mongo DB

Course Outcomes:

After the successful completion of the course, the students will be able to

CO1 Develop React applications using React basic library, forms and routing

CO2 Develop Node.js applications using library node.js

CO3 Develop Client Server applications using Express.js framework

CO4 Construct Node.js applications using Mongo DB databases.

Mapping of Course Outcomes with POs and Program Specific Outcomes(PSOs):

CO	Program Outcomes(POs)												PSOs		
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
CO 1	3	3	3	3	3	3	-	-	3	2	-	3	3	3	3
CO 2	3	3	3	3	3	3	-	-	3	2	-	3	3	3	3
CO 3	3	3	3	3	3	3	-	-	3	2	-	3	3	3	3
CO 4	3	3	3	3	3	3	-	-	3	2	-	3	3	3	3

UNIT - I **(18 Hours)**

React.js: MERN, MERN Components, Serverless Hello World Application, ES6, DOM, Virtual DOM, Installation. Components in React, using JSX, React Project Structure, create-react-app, Props, State, component life cycle, Handling Events, Component Communication, Data Binding – One way, two way, SPA.

Working with Forms & Third Party libraries, Routing.

List of Experiments

1. Develop a Calculator React Application
2. Develop a News feed application
3. Develop React application using Forms
4. Develop a website to demonstrate React Routing.

UNIT - II **(18 Hours)**

React.js & Node.JS: Redux Application Architecture, Integrating Redux with React, Testing & Deployment of React Applications.

Node.js, Using Events, Listeners, Timers, and Callbacks in Node.js, Handling Data I/O in Node.js. Accessing the File System from Node.js, Implementing HTTP Services in Node.js.

List of Experiments

1. Develop a Redux application
2. Testing & Deploying React application to GitHub.
3. Develop a Node.js application demonstrating handling data I/O (Buffer, Stream, Zlib modules)
4. Demonstrate accessing Filesystem from Node.js application
5. Implement a HTTP Client and HTTP Server.

UNIT - III **(18 Hours)**

Express.js: Routes, Request and Response objects, Template engine. Understanding middleware, Query middleware, Serving static files, Handling POST body data, Cookies, Sessions, Authentication.

List of Experiments

1. Demonstrate Express Routing.
2. Demonstrate Express.js Cookies
3. Demonstrate Express.js Sessions
4. Demonstrate Express.js Authentication

UNIT - IV **(18 Hours)**

MongoDB: Understanding NoSQL and MongoDB, using mongoDB Compass, using MongoDB Shell, Accessing MongoDB from Node.js.

List of Experiments

1. Demonstrate working with MongoDB using MongoDB Compass.
2. Demonstrate working with MongoDB using MongoDB Shell.
3. Demonstrate accessing MongoDB from Node.js

TEXT BOOKS:

1. Anthony Accomazzo, Ari Lerner, Clay Allsopp, David Guttman, Tyler McGinnis, and Nate Murray. *Fullstack React - The Complete Guide to ReactJS and Friends*. Fullstack.io, 1 edition, 2017. ISBN 9780991344628

2. Brad Dayley, Brendan Dayley, and Caleb Dayley. *Node.js, MongoDB and Angular Web Development*. Addison-Wesley, 2 edition, 2018. ISBN 9780134655536

REFERENCES:

1. Mark Tielens Thomas. *React in Action*. Manning, 1 edition, 2018. ISBN 9781617293856

2. Alex Young, Bradley Meck, Mike Cantelon, Tim Oxley, Marc Harter, T.J. Holowaychuk, and Nathan Rajlich. *Node.js in Action*. Manning, 2 edition, 2017. ISBN 9789386052049

3. Kyle Banker, Peter Bakkum, Shaun Verch, Douglas Garrett, and Tim Hawkins. *MongoDB in Action*. Manning, 2 edition, 2016. ISBN 9789351199359

4. Vasan Subramanian. *Pro MERN Stack*. Apress, 2 edition, 2019. ISBN 9781484243909